Enrollment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Computer Application
BCA / B.Sc. (IT) | Sem-5 | Winter-2025

Course Code	:	2304CS501	**Date**	:	15-10-2025
Course Name	:	Introduction to Advanced Technologies	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Explain the characteristics of Cloud Computing.	**4**
	(B)	List different types of cloud service models and Explain any one in detail.	**3**
		OR	
		What is Virtualization? Explain its role in cloud computing.	
	(C)	Explain in detail different types of cloud with neat diagram.	**7**
		OR	
		Explain the Cloud Computing Architecture and Technologies in detail.	
Q.2	**(A)**	Explain the Technology and Features of Augmented Reality (AR) and Virtual Reality (VR).	**4**
	(B)	List any three challenges with AR and VR.	**3**
		OR	
		Write any three real-life examples where AR and VR is used.	
	(C)	Explain various Applications of AR and VR in detail.	**7**
		OR	
		Explain AR Systems and their Functionality. Also describe visualization techniques used in AR.	
Q.3	**(A)**	Explain the different types of Machine Learning.	**4**
	(B)	List any three Tools or Technologies commonly used for Machine Learning.	**3**
		OR	
		What are the main components of a Machine Learning Model.	
	(C)	Explain the Machine Learning Process / Task in detail with a neat diagram.	**7**
		OR	
		Explain Artificial Neural Networks (ANN) with structure and working.	

Q.4	**(A)**	Explain the Data Science Process in detail.	**4**
	(B)	What are the different roles in Data Science?	**3**
		OR	
		List any three Python libraries used in Data Science and their uses.	
	(C)	Explain different Data Operations in Data Science with examples.	**7**
		OR	
		Explain the Need and Applications of Data Science in detail.	
Q.5	**(A)**	Explain the types of Blockchain in detail.	**4**
	(B)	What is the need for Blockchain technology?	**3**
		OR	
		List any three important Blockchain terminologies and explain briefly.	
	(C)	Explain the Blockchain Architecture with a neat diagram.	**7**
		OR	
		Explain different Consensus Algorithms in Blockchain.	

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